FORM HS-02 · REV. 5 · POST-STASIS REFRESHER · MODULE 2 OF 9 · POST AT EVERY PPE ISSUE POINT

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

BREATHE CLEAN.

THE SAND UNDER YOUR BOOTS IS GROUND GLASS FOR YOUR LUNGS. DUST YOU CAN'T SEE DOES THE DAMAGE.

THE SAND IS SILICA

SILICOSIS HAS NO CURE.

This world's sand is rich in crystalline silica. Cutting, grinding, drilling, blasting or just driving over it throws up **respirable dust**, particles small enough to reach deep into your lungs. Your lungs scar around them, and the scarring never heals. It keeps getting worse after exposure stops.

SILICOSIS

Scarred, stiff lungs. Shortness of breath that never goes away. Can kill.

LUNG CANCER

Crystalline silica is a known human carcinogen.

COPD

Chronic bronchitis and emphysema. Every breath is work.

KIDNEY DISEASE

Silica exposure is linked to kidney damage. It also raises your risk of TB.

THE LIMITS

RESPIRABLE CRYSTALLINE SILICA
8-HOUR AVERAGE

50 µg/m³
PERMISSIBLE EXPOSURE LIMIT (PEL)

25 µg/m³
ACTION LEVEL: MORE SAMPLING

Invisible
RESPIRABLE SILICA IS TOO FINE TO SEE.
VISIBLE DUST MEANS FAR MORE IS THERE.

AIR SAMPLING

- **Industrial Hygiene** samples dust at active work faces on the schedule in the written dust control plan.
- You get your result in writing, or posted at muster, within 5 working days.
- **At or above 25 µg/m³:** the area is sampled again within 6 months.
- **Above 50 µg/m³:** the controls are fixed, respirators are assigned, and the area is sampled again within 3 months.
- Tasks done exactly by the Old Earth task list (Table 1) use that list's controls and respirators instead.

SILICA MEDICAL CHECKS

- **Who:** anyone who must wear a respirator for silica on 30 or more days a year. Site rule: anyone sampled at or above 25 µg/m³ on 30 or more days a year too.
- **When:** a free exam within 30 days of the assignment, then at least every 3 years.
- **What:** work and medical history, a physical exam focused on your lungs, a chest X-ray, a lung function test, and a TB test at the first exam.
- Industrial Hygiene puts you on the list and books your exam at Central Med.
- A cough or shortness of breath that won't go away: go to the Forward Aid Station (FAS).

CONTROLS FIRST

RESPIRATORS ARE THE LAST LINE OF DEFENSE

1. **Wet it.** Cut, grind, core and drill with water fed to the blade or bit. Wet sweep, never dry sweep. Never clear dust with compressed air. Clean up wet slurry before it dries: dried slurry is dust again.
2. **Vacuum it.** Tools with shrouds connected to a HEPA (high-efficiency particulate air) dust collector. Clean up with a HEPA vacuum.
3. **Pull it away.** Local exhaust at fixed cutting and grinding stations in the Fabrication Yards.
4. **Enclose yourself.** Equipment cabs sealed, filtered, pressurized and air-conditioned. Keep the doors shut. Check your cab every shift: door seals closed, pressure gauge in the green, filter in date, AC working, floor clear of dust.
5. **Then wear the respirator** the dust control plan assigns. See the ladder below.

Who decides: every dust task runs under a written dust control plan. The plan names a competent person for silica (badge endorsement) who checks the controls at the start of every shift and can stop the work. Industrial Hygiene picks the respirator for each task from the plan or the latest reading. Never pick a lower rung yourself.

RESPIRATOR LADDER

ASSIGNED PROTECTION FACTOR (APF)
HIGHER = CLEANER AIR INSIDE

N95 filtering facepiece Disposable. Short, light dust tasks the plan allows.	10 × CLEANER
Half-mask elastomeric P100 cartridges. Routine cutting and grinding.	10 × CLEANER
Loose-fitting PAPR Powered air-purifying respirator with a hood. Works with beards. Cool airflow.	25 × CLEANER
Full-facepiece elastomeric P100. Heavy dust, protects eyes too.	50 × CLEANER
Tight-fitting full-face PAPR Heaviest dust, long tasks, rescue work in blowing sand.	1000 × CLEANER
Abrasive blasting Not on this ladder	NIOSH-approved Type CE supplied-air blasting hood on a clean Grade D breathing-air line. Never a PAPR or a cartridge mask.

A respirator adds breathing strain and heat. Take your breaks in clean, cool air with the mask off, and tell your crew lead if you feel short of breath. Work/rest follows the table in Module 3 (HS-03).

DO

- **Medical evaluation first.** Revival Med Bay must clear you to wear a respirator before your fit test. Tell the FAS if breathing gets hard in the mask.
- **Get fit-tested** by Industrial Hygiene on your exact model and size. Repeat every year, and after any change to your face or a big change in weight (revived workers gain weight back).
- **Seal check every time.** Negative: cover the cartridges and breathe in gently. The mask pulls in slightly and stays in for about 10 seconds. Positive: cover the exhale valve and breathe out gently. The mask bulges slightly with no leaks. N95: cup both hands over it, breathe out hard and feel for air at the edges.
- Check fails? Re-seat the straps and try again. Still failing: stay out of the dust and see Industrial Hygiene for a refit.
- **Be clean-shaven** where the seal touches your face. Beards go under a loose PAPR hood only.
- **Inspect it before every use:** straps, valves, cracks.
- Change filters when breathing gets hard, when they're damaged, wet or dirty, or on the schedule Industrial Hygiene sets. PAPR: change the filter when the flow alarm sounds.
- **Mask fails in the dust** (you taste or smell dust, it's hard to breathe, it leaks, or the PAPR alarm sounds): walk out of the dust at once, then fix or swap it.

DON'T

- Wear a mask before your medical evaluation, or one you weren't fit-tested on.
- Wear a tight-fitting mask over stubble or a beard.
- Take it off inside the dust, even to fix it.
- Hang it around your neck between cuts. The inside fills with dust.
- Use a particle respirator (N95 or P100) for gases, vapors or low oxygen. Gases and vapors need the cartridge Industrial Hygiene assigns (Module 5, Module 8). Low oxygen needs supplied air (Supplement A (HS-SA)).

RESPIRATOR PROGRAM

Industrial Hygiene runs the respiratory protection program.

- Order: medical evaluation at Revival Med Bay, then training, then your fit test. Fit test and training again every year.
- Industrial Hygiene picks every respirator and cartridge, and sets the cartridge change schedule.
- Supplied air is Grade D breathing air. Compressor intakes sit upwind, away from sand, engine exhaust and vents.
- PAPR batteries: charge every shift and keep them out of the sun.
- Clean your respirator after every shift. Store it sealed, out of the sun and sand.

PPE CARE

- **Helmet:** check it every shift for cracks, dents, a chalky or faded shell, and a worn suspension. Sun and heat break shells down fast. Replace it after any hit, and by the date on the label. Never leave it on a dashboard or in the sun. Issued helmets carry the HT (high temperature) mark. Only the issued RAMP band and labels go on it.
- **Eyewear** pitted or scratched by sand: swap it.
- **Gloves:** the right glove for the hazard (cut, heat, or chemical per the SDS). Swap worn or soaked gloves.
- **Hi-vis** that's faded or dirty, or has cracked reflective strips, isn't hi-vis. Swap it.
- **FR clothing:** no holes, no oil or solvent soaking. **Cooling vest:** recharge or swap it at every break.
- Free exchange at the Quartermaster (Ch 15), any shift.

EYES

- Safety glasses marked **Z87+** (the + means high impact) at all times on site.
- Outdoors in daylight, wear tinted lenses with UV protection: look for **U6** (the top UV rating) next to Z87+. Clear lenses at night.
- **Sealed goggles** marked D4 (dust) or D5 (fine dust) whenever sand is blowing, when cutting or grinding, and outside a cab in wind of 15 mph or more. Chemical splash goggles are marked D3.
- Face shield **over** glasses or goggles for grinding and chemical pours. A shield alone isn't eye protection.
- Sand in your eye: don't rub. Flush at an eyewash station until it's clear, then go to the FAS. Chemical in the eye: keep flushing at the station for the full 15 minutes. The medic comes to you.
- **Eyewash** is within 10 seconds of every chemical hazard, activated weekly and inspected yearly. Units are shaded and insulated or chilled so the water stays lukewarm, 60–100°F (site rule). Never flush with water that feels hot. Hot or dry station: tell your crew lead at once and use the nearest other one.

HEARING

85
DBA SITE LIMIT, 8-HR AVERAGE (SITE RULE)

3 dB
EVERY 3 DB LOUDER HALVES YOUR TIME

NOISE	85	88	91	94	97	100 DBA
Max time	8 hr	4 hr	2 hr	1 hr	30 min	15 min

Shifts longer than 8 hours get lower limits, posted by Industrial Hygiene. Old Earth law: 90 dBA for 8 hours, halving every 5 dB, with a hearing program from 85 dBA. dBA = decibels, A-weighted: loudness as the ear hears it.

- **Rule of thumb:** if you have to shout to be heard at arm's length, you need protection.
- Plugs rolled tight and fully in, or muffs sealed over your ears. **Double up** (plugs and muffs) above 100 dBA and near chipping guns, generators and impact tools. Then watch for strobes and keep Ch 1 loud enough to hear.
- **Hearing program (Industrial Hygiene):** noise mapping and posted "Hearing protection required" zones from 85 dBA. A baseline hearing test before your first noisy shift (after 14 hours away from loud noise), then a test every year. If a test shows your hearing has shifted, you're told in writing within 21 days and refitted.
- The Quartermaster issues plugs and muffs and shows you how to fit them. Ask for another type if they don't seal.
- Hearing loss from noise is permanent. Ringing ears after a shift means damage already happened.

PPE BY TRADE

● ALWAYS · ○ FOR THE LISTED TASKS

Everyone on site: Z87+ glasses at all times. Helmet, hi-vis (Class 2 by day, Class 3 at night, or a hi-vis cooling vest) and safety boots outside cabs, shelters and enclosed bays.

TRADE	HELMET	Z87+ EYES	GOGGLES / SHIELD	RESPIRATOR	HEARING	GLOVES	HI-VIS	OTHER
Welder	●	●	●	●	○	●	●	Welding helmet, FR leathers. Hi-vis must be flame-resistant (Module 5).
Machinist	○	●	○	○	●	○	○	Helmet and hi-vis whenever you leave the machine bay or work under cranes or overhead work. No gloves at rotating spindles. No loose sleeves or jewelry (Module 7).
Equipment operator	●	●	○	○	●	●	●	Sealed cab, checked every shift. Carry your fitted respirator in the cab. Filter, pressure or AC fails: respirator on, park, call your crew lead.
Rigger / signal	●	●	○	○	○	●	●	Orange helmet (Module 6).
Polymer tech	●	●	●	●	○	●	●	Chemical gloves and suit per the SDS (Safety Data Sheet) (Module 8).
Laborer	●	●	○	○	○	●	●	No dry sweeping: wet sweep or HEPA vacuum. Demolition and cleanup: the respirator the dust control plan assigns. An N95 only where the plan says APF 10 is enough.
Electrician	●	●	○	○	○	●	●	Class E helmet (blue). Arc-rated clothing and voltage-rated gloves per Supplement B (HS-SB).

Boots: safety-toe, puncture-resistant, heat-rated soles for everyone. Sand surfaces pass 150°F by midday. Shake out boots, gloves and PPE before you put them on: sand stingers hide in stored gear.

WIND AND BLOWING DUST

SITE WIND TABLE · SAME ON EVERY MODULE

WIND	RULE
15+ mph	Goggles outside cabs. Secure sheet goods and loose material.
20+ mph sustained or gust	No outdoor hot work. No crane personnel platforms. High-sail loads need Lift Director approval.
25 mph sustained or 30 mph gust	Off open steel, roofs, leading edges, swing stages and outdoor scaffolds above Deck 1. MEWPs down (or the maker's limit if lower).
30 mph sustained or gust	All crane loads landed. Booms secured per maker.
Blowing-dust advisory (visibility 1 mile or less)	No outdoor hot work. Lifts stop whenever the operator loses sight of the load or signal person.

Wind = the higher of your local anemometer (crane, MEWP or crew meter) and the Ch 16 posting. Gust = 3-second gust. These are DCW limits; several are stricter than Old Earth rules. Follow the maker's limit if it's lower.

SANDSTORM

Atmos calls it on Ch 16 and Hub Command rebroadcasts it on Ch 1. Visibility can drop to zero in minutes. Wind-driven sand strips skin and chokes filters.

- **Watch:** a storm is possible within 2 hours. Secure loose material, plan how loads come down, stage crews near shelter.
- **Warning:** arrival expected within 30 minutes. Land all loads, booms down (tower cranes to free-slew per maker), climbers and MEWPs down, move to shelter.
- **Take-shelter siren:** steady tone and red strobes, no later than 15 minutes before arrival.

**MAKE SAFE.
GET SEALED IN.**

Every radio scans Ch 1, Ch 16 and your sector channel.

1. **Make your work safe:** torch off and cylinder valves closed, load landed, machine parked with attachments down.
2. **Goggles and respirator on** before you move. Full-face if you have it.
3. **Go to the nearest sealed shelter** with your buddy. Every grid block has one. Don't try to reach the muster point.
4. **Operators and drivers:** when the siren sounds, stop. No machine moves while people walk to shelters. No shelter close and in sight? Stay in your sealed cab: brake set, attachments down, beacon on, AC and pressurization running. Radio your location to your crew lead.
5. Caught in the open? **Get low behind solid cover**, back to the wind, face covered. Stay put. Ch 1: "Emergency, emergency, emergency. Sandstorm," then your location, how many of you, any injuries and your call sign.
6. **In the shelter:** the senior person counts heads and reports as "Shelter [Sector]-[grid]" on that shelter's sector channel. Missing persons go on Ch 1. Shelter full: the senior person radios to send you to the next one. Cooling or seal fails: report it on Ch 1 at once and stay put unless Fire & Rescue orders a move.
7. **Leave only on the Ch 1 voice all clear** ("All clear, all clear, [area]") or your crew lead. Never on a tone alone.

STOP-WORK AUTHORITY

Anyone can stop unsafe work. No penalty, no questions. Stop work when dust is visible in the air, a dust collector is down, or someone has no respirator for the job. Stop it, make it safe, tell your crew lead (your foreman if there's no crew lead; any green hat if it isn't fixed). Work restarts only when the crew lead confirms the fix with the person who stopped it. Every stop is logged.

WORKER ACKNOWLEDGEMENT

I know I need a medical evaluation and a fit test before I wear a respirator, and how to seal-check it. I know the silica controls, the hearing rules and how to care for my PPE. I know the wind table, the sandstorm procedure and my nearest shelter.

NAME
COLONIST ID
CREW / SECTOR
DATE

Adapted from Old Earth standards: OSHA 29 CFR 1926.1153 respirable crystalline silica (exposure assessment, medical surveillance, Table 1), OSHA 1910.134 respiratory protection (APFs, medical evaluation, fit testing, Appendix B-1 user seal checks, Grade D air), OSHA 1926.57(f) and 1910.94(a) abrasive blasting, CGA G-7.1 breathing air, NIOSH Criteria for Occupational Noise Exposure (1998: REL 85 dBA, 3 dB exchange), OSHA 1926.52 and 1910.95 noise and hearing conservation, ANSI/ISEA Z87.1-2020 eye protection, ANSI/ISEA Z89.1-2014 head protection, ANSI/ISEA 107-2020 high-visibility apparel, ANSI/ISEA Z358.1-2014 emergency eyewash, NWS dust storm safety.